Car Rider Pickup System for Schools

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The Wranglers

Problem Background

- Over 70% of students are now picked up via car, creating massive congestion.
- Parents wait up to 1 hour, with lines often forming 1-2 hours before dismissal.
- Queues spill onto local highways, causing traffic backups and increasing accident risks.
- Inclement weather increases delays as bus and bike riders often switch to the car line.
- The manual process requires 5-10 staff members daily, diverting them from educational duties.

Problem - Needs Statement

School pickup via the car rider line has been increasing in the US over the years leading to an increase in line length and pickup time. Now, over 70% of students are picked up from school via the car rider line. Car rider lines have become congested, posing a safety issue for both the students and other drivers. With more students attending school, car rider lines have become too slow and cause parents to wait up to over an hour in line leading to frustration. Some studies show that the car rider pick-up line starts to form 1-2 hours before school actually gets out, which leads to traffic backing up local highways and streets, and even frequent accidents. Inclement weather further slows the line, as students who ride the bus or bike to school will often be picked up by their parents in the car rider line to avoid the weather. Additionally, 5-10 staff members are required to set up and run the pickup process, diverting their time away from the education of students. An automated system is needed to streamline the pick-up process to decrease delays and safety issues.

Goal and Objectives

Goal

The goal of this project is to design and prototype a system that will help automate the car rider pick up process at schools. The system is built to reduce the amount of time necessary to pick up students from school.

Objectives

- The system will allow parents to update their information (license plate numbers, approved adults for pickup, car rider schedules) via a website with approval from a staff member of the school.
- The system will read the marker on the car (license plate, QR code, nametag) as it enters the pickup line and send the information to a server.
- The server will identify the parent and child associated with the car marker, and update a display at the entrance of the pickup line to direct the car to the proper station.
- The laptop at the front of the school is updated to inform the students and staff which child is being picked up, and which zone to go to.

Goal and Objectives - Requirements

Marketing Requirements

1. Make the pickup process faster for parents
2. The system should be easily updated
3. The system should be low cost
4. The system should be easy to use/install
5. Information must be validated if updated or inserted
6. Student safety should be ensured (information, physical, etc.)
7. The system should be reliable

Goal and Objectives - Requirements

Marketing Requirements	Engineering Requirements
1	Pickup time should be reduced by 30% per 100 kids. The time from the car entering the view of the camera to information being displayed should be no more than 5 seconds.
4	7 inch (diagonal) display screen to give station and process instructions clearly.
1, 6	Per Texas Family Code, ensure that students are referred to authorized parents 100% of the time.
6	Private Information stored in the database should be encrypted using 128 bit encryption.
7	Should have a stable connection from within 1 kilometer, stable power to all components.
1,6,7	Should correctly identify/assign parents to kids 100% of the time.
3	The costs for developing the system should not exceed $300.
2,4,5	The system should allow parents to submit a request to the school to update information and the school should receive the request all within 1 minute.

Goal and Objectives - Requirements

Marketing Requirements	Engineering Requirements
4	The system should have a modular design such that failed components can be replaced by a technician in under 15 minutes.
4,7	The system should be mounted correctly on a tripod to ensure stability and prevent hazards. Sandbag/Weights of at least 1 pound to keep it grounded.
7	The software will comply with the Digital Millennium Copyright Act.
2,4,5,6	Staff should be able to access databases with provided credentials and view all parent to children links.
4,7	The system should be able to withstand 20mph wind.
1	Minimum 40inch (diagonal) display screen to give station and process instructions clearly inside in the pickup .
4	The system should have a modular design such that failed components can be replaced by a technician in under 15 minutes.
4,7	The system should be mounted correctly on a tripod to ensure stability and prevent hazards. Sandbag/Weights of at least 1 pound to keep it grounded.

Design Constraints and Feasibility

Technical and Temporal Constraints

- Maximum of 5 seconds from image capture to display update
- Identification accuracy must be 100% to ensure safety and efficiency
- Connection between the system parts must be reliable

Physical and Environmental

- Needs to withstand outdoor elements (wind and rain)
- Needs to be portable and stable with modular parts, mounting, and power supply to reduce system downtime and speed up repair and maintenance

Economic

- Must cost $250 or less, necessitating cost-effective components

Feasibility

- Core technology, such as Python image processing and Wi-Fi, is well documented and achievable
- Main challenge will be creating a fast and efficient system within our budget

Evaluation of alternative solutions and Rationale

Solution	Pros	Cons
Manual Pickup App (PikMyKid)	Easy to implement; low cost; reduces paperwork	Requires staff queue management; no automated vehicle identification
Integrated Dismissal System (Beeline)	Improves communication; integrates with school systems	Requires training and infrastructure; still staff-monitored
Geofencing (Car-Pool App)	Real-time arrival alerts; reduced manual check-ins	Dependent on GPS accuracy, mobile devices, and connectivity
GPS Check-In (IschoolRide)	Keeps students inside until arrival; reduces idle staging	Relies on parent app usage and accurate GPS; staff verification will still be needed
QR Codes (Kids Pick Up App)	Simple, affordable; no cellular service required	Requires staff scanning; affected by glare, weather, and visibility

All alternatives improve organization but still rely on staff actions or parent interaction, limiting full automation.

System Level Functional Description - High-Level Block Diagram

- Recognize car identifier (license plate or QR code)
- Send and receive data
  - Over zigbee
- Instruct parents what zone to go to

Car Entrance Module

Car line entrance

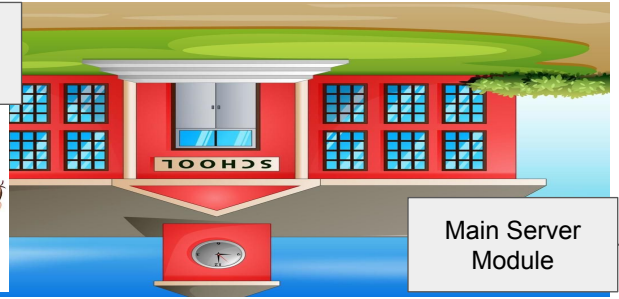
Zone Pickup Module

- Display pickup information to staff
- Allow staff to update system once a kid has been picked up
- Forward data between car entrance module and main server module

Car Stations

- Display website which contains instructions to send kid to car stations

Kids Waiting Module



Main Server Module

- Host a website that can be updated in real time
- Connect to a database that contains parent/kid info
- Handle information update requests from parents
- Send updates to car entrance module

System Component - Car Entrance Module

Hardware:

- Microcontroller for processing and controlling devices
- Camera for images
- Display for instructing parents
- Zigbee antenna for radio communication
- Rechargeable battery for power

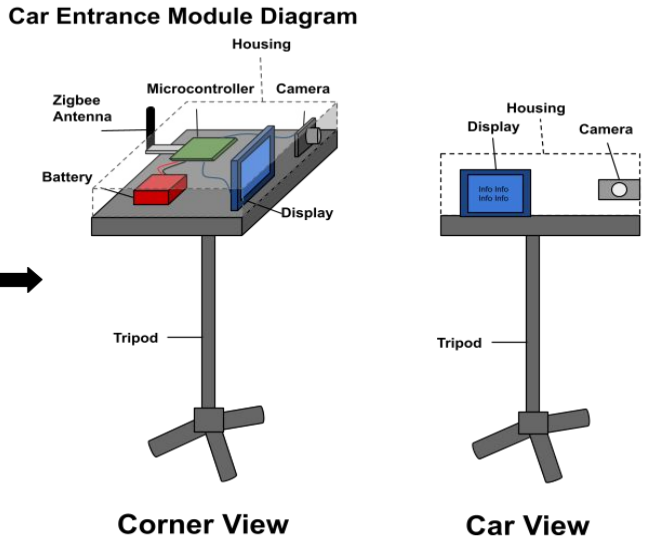
Software:

- Image processing and recognition software
- Display instructions on screen
- Zigbee communication software

Interfaces:

- Inputs:
  - Images
  - Zigbee data
- Outputs:
  - Display information to instruct parents
  - Zigbee data

Car Entrance Module



System Component - Zone Pickup Module

Hardware:

- Laptop for software and zigbee connection
- Zigbee antenna for radio communication
- Phones to show specific zone information to zone attendant

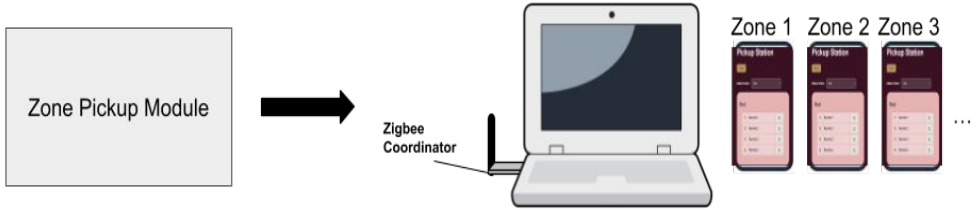
Software:

- Zigbee communication software
- Website access on phones

Interfaces:

- Inputs:
  - Zigbee data
  - Button input on phones once a kid has been picked up
  - Server data (updates for car entrance module)
- Outputs:
  - Zigbee data

Zone Pickup Module Diagram



System Component - Main Server Module

Hardware:

- Computer to run software

Software:

- Website hosting - needs real time updating
- Database connection to access and update parents/kids info
- Send updates to car entrance module display

Interfaces:

- Inputs:
  - Identifier data (to query database to find parent and kid)
  - Parent requests (to update information)
- Outputs:
  - Information to update car entrance module display

Main Server Module Diagram

System Component - Kids Waiting Module

Hardware:

- Large display (projector or large TV)

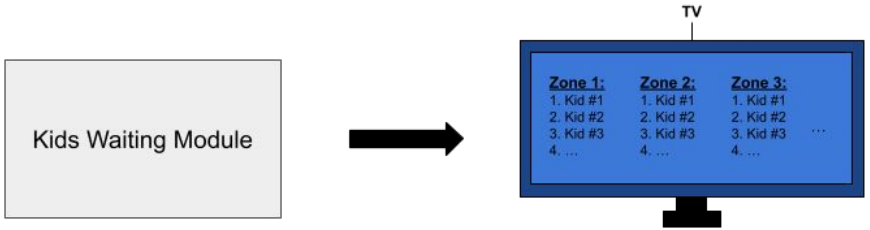
Software:

- Website access on screen

Interfaces:

- Inputs: None
- Outputs:
  - Website to instruct kids and staff when to send kid to zones to be picked up (also which zone to go to)

Kids Waiting Module Diagram



Design Validation and Testing - 1

Small, foundational testing first

- Verify connectivity to web server and database
- Display text found by the camera directly to the screen it is connected to
- Ping router from the car entrance module to verify connectivity
- QUERY database to ensure proper initialization

System testing comes next

- Packets should be sent to the database, should return student names
- Screens directing drivers and students should update in no more than 5 seconds
- Web server flags incoming requests to alter database to admin page

Design Validation and Testing - 2

End with Full System Functionality Tests:

- Run full pickup simulation
  - Scan QR code
  - Send information to Server
  - Inform kid they are being picked up
  - Kid goes to the line
  - Kid gets in the car

Vision for Product Demonstration

- Film a video of the full system test
  - Set up each of the modules in locations similar to what the school would look like and go through the process of picking up a queue of students
- Live QR code scanning and site updates
- Website - https://thewranglers.onrender.com/

Evidence of Effective Teaming

Defined Roles and Responsibilities

Each team member is responsible for a specific subsystem:

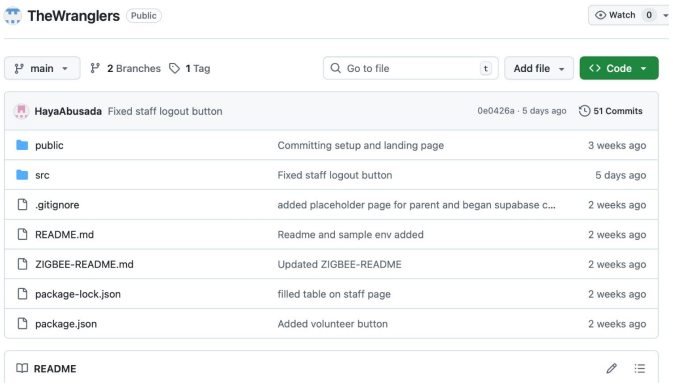
- **Haya Abusada** – Frontend development and system integration
- **Sara Vorhaben** – Backend development and authentication
- **Jake Hewett** – Database architecture and administrative interface
- **Jack Uehlinger** – Hardware and camera integration
- **Jayden Pearson** – Mechanical design and enclosure development

This ensures **clear ownership of tasks and efficient development**.

Collaboration and Communication

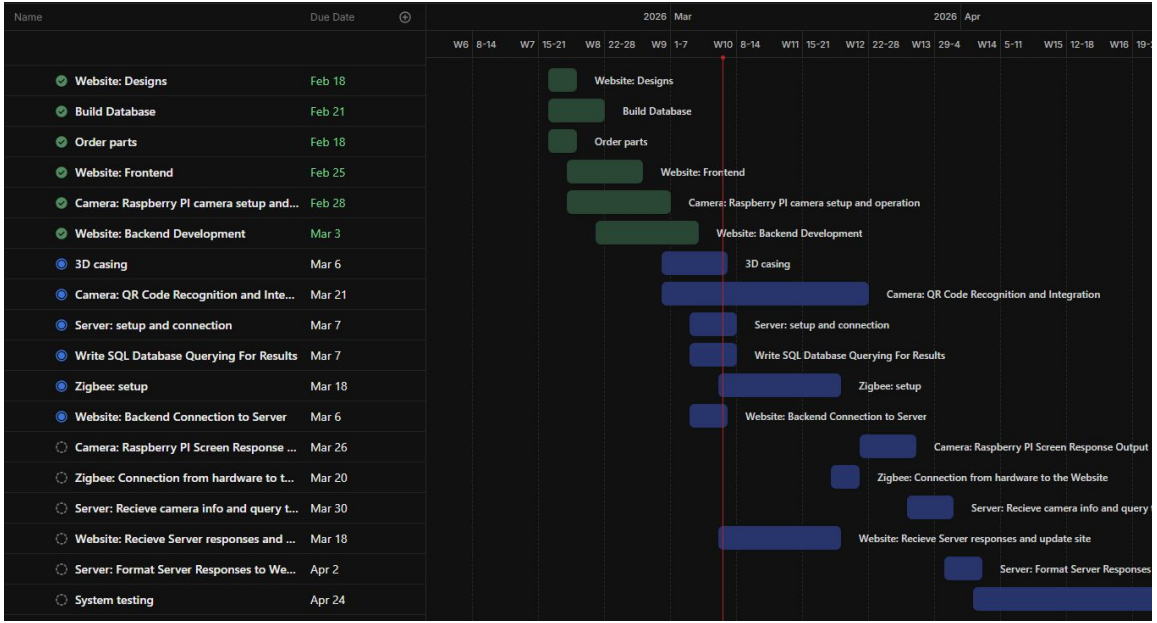
The team maintains strong coordination through:

- **Weekly team meetings** to review progress and address issues
- **GitHub** for version control and collaborative development
- **Discord** for daily communication and troubleshooting
- **Google Drive** for shared documentation and reports



Screenshot of Github Repo

Evidence of Effective Teaming - Gantt Chart



- Gantt Chart Shows tasks completed
- Red line is today
- All tasks are completed on time

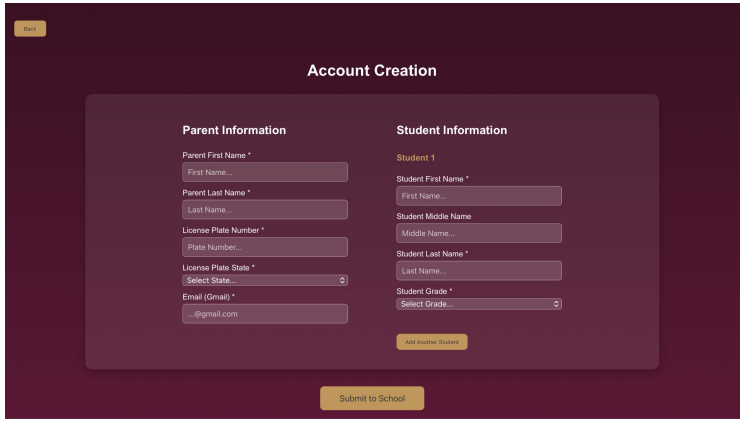
Preliminary Results

System Development Progress

- Web-based interface developed using **React**
- Major pages implemented:
  - Landing page
  - Parent portal
  - Administrative pages
  - Kids pickup module
- Website successfully deployed and connected to **database.**

Parent Portal Functionality

- **Google authentication** implemented for secure login
- Parents can:
  - Register student information
  - Update license plate details
  - View student pickup status
- Website successfully **reads and writes data to the database**

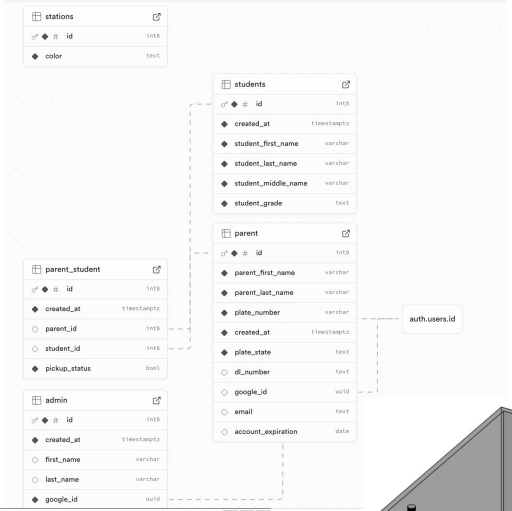


Account creation page of the website to enter required info and create new accounts.

Preliminary Results

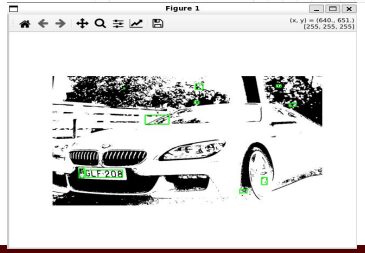
Database Implementation

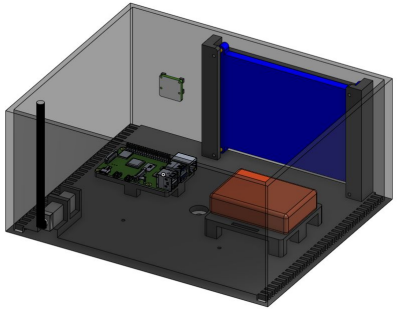
- PostgreSQL database created using **Supabase**
- Tables implemented for:
  - Parent accounts
  - Student records
  - Parent–student relationships
  - Pickup status tracking
- System successfully **inserts, retrieves, and updates records**



Hardware Progress

- **Raspberry Pi configured and connected via SSH**
- Camera module installed and tested
- System can **process images and detect license plates**





Mechanical Development

- CAD models created for device housing
- Measurements completed for hardware mounting
- Prototype enclosure design in progress

Questions?

Thanks for listening!

The Wranglers